Luleå Technical University
School of Computer Science, Electrical and Space Engineering
Luleå, Sweden

Lab
Advanced deep learning D7047E 47620 VT2026

LAB0 THEORY

Ekpenyong Esu Ekpenyong
ekpekp-5@student.ltu.se

Sowmya Nelakurthi
sownel-4@student.ltu.se

Sreenidhi Thallam
sretha-5@student.ltu.se

Examiner: Marcus Liwicki
LTU, Luleå

29th March 2026

Part 1: Theoretical Tasks

Task 0.1 — Convolution

Setup

We are given image $\mathbf{I}$ and kernel $\mathbf{k}$:

$$\mathbf{I} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & -3 & -4 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \qquad \mathbf{k} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

A **same convolution** with zero padding and stride $(1, 1)$ produces an output of the same spatial size as the input (4×4). We pad $\mathbf{I}$ with a border of zeros:

$$\mathbf{I}_{\text{padded}} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 2 & 1 & 0 \\ 0 & 1 & -3 & -4 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

For each output position (r, c), we extract the 3×3 patch centered at that position from the padded image, multiply element-wise with $\mathbf{k}$, and sum:

$$(\mathbf{I} * \mathbf{k})_{r,c} = \sum_{i=0}^{2} \sum_{j=0}^{2} k_{i,j} \cdot I_{\text{padded}}[r + i,\ c + j]$$

Calculation

For each output position (r, c), the 3×3 patch is taken from padded rows $[r - 1,\ r,\ r + 1]$ and padded cols $[c - 1,\ c,\ c + 1]$ (using 1-based indexing for the original image).

Position $(1, 1)$ — padded rows 0–2, cols 0–2:

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \odot \mathbf{k} = 0 + 0 + 0 + 0 + 2 + 1 + 0 + 1 + 0 = \mathbf{4}$$

Position $(1, 2)$ — padded rows 0–2, cols 1–3:

$$\begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} \odot \mathbf{k} = 0 + 0 + 0 + 1 + 2 + 1 + 0 + 1 + 0 = \mathbf{5}$$

Position $(1, 3)$ — padded rows 0–2, cols 2–4:

$$\begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix} \odot \mathbf{k} = 0 + 0 + 0 + 1 + 2 + 1 + 0 + 2 + 0 = \mathbf{6}$$

Position $(1, 4)$ — padded rows 0–2, cols 3–5:

$$\begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 1 & 0 \end{pmatrix} \odot \mathbf{k} = 0 + 0 + 0 + 1 + 2 + 0 + 0 + 1 + 0 = \mathbf{4}$$

Position $(2,1)$ — padded rows 1–3, cols 0–2:

$$\begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & -3 \end{pmatrix} \odot \mathbf{k} = 0+1+0+0+2+1+0+1+0 = \mathbf{5}$$

Position $(2,2)$ — padded rows 1–3, cols 1–3:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & -3 & -4 \end{pmatrix} \odot \mathbf{k} = 0+1+0+1+2+2+0-3+0 = \mathbf{3}$$

Position $(2,3)$ — padded rows 1–3, cols 2–4:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ -3 & -4 & 1 \end{pmatrix} \odot \mathbf{k} = 0+1+0+1+4+1+0-4+0 = \mathbf{3}$$

Position $(2,4)$ — padded rows 1–3, cols 3–5:

$$\begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 0 \\ -4 & 1 & 0 \end{pmatrix} \odot \mathbf{k} = 0+1+0+2+2+0+0+1+0 = \mathbf{6}$$

Position $(3,1)$ — padded rows 2–4, cols 0–2:

$$\begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & -3 \\ 0 & 1 & 1 \end{pmatrix} \odot \mathbf{k} = 0+1+0+0+2-3+0+1+0 = \mathbf{1}$$

Position $(3,2)$ — padded rows 2–4, cols 1–3:

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & -3 & -4 \\ 1 & 1 & 1 \end{pmatrix} \odot \mathbf{k} = 0+1+0+1-6-4+0+1+0 = \mathbf{-7}$$

Position $(3,3)$ — padded rows 2–4, cols 2–4:

$$\begin{pmatrix} 1 & 2 & 1 \\ -3 & -4 & 1 \\ 1 & 1 & 1 \end{pmatrix} \odot \mathbf{k} = 0+2+0-3-8+1+0+1+0 = \mathbf{-7}$$

Position $(3,4)$ — padded rows 2–4, cols 3–5:

$$\begin{pmatrix} 2 & 1 & 0 \\ -4 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \odot \mathbf{k} = 0+1+0-4+2+0+0+1+0 = \mathbf{0}$$

Position $(4,1)$ — padded rows 3–5, cols 0–2:

$$\begin{pmatrix} 0 & 1 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \odot \mathbf{k} = 0+1+0+0+2+1+0+0+0 = \mathbf{4}$$

Position $(4,2)$ — padded rows 3–5, cols 1–3:

$$\begin{pmatrix} 1 & -3 & -4 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \odot \mathbf{k} = 0-3+0+1+2+1+0+0+0 = \mathbf{1}$$

Position $(4,3)$ — padded rows 3–5, cols 2–4:

$$\begin{pmatrix} -3 & -4 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \odot \mathbf{k} = 0 - 4 + 0 + 1 + 2 + 1 + 0 + 0 + 0 = \mathbf{0}$$

Position $(4,4)$ — padded rows 3–5, cols 3–5:

$$\begin{pmatrix} -4 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \odot \mathbf{k} = 0 + 1 + 0 + 1 + 2 + 0 + 0 + 0 + 0 = \mathbf{4}$$

Result

$$\boxed{\mathbf{I} * \mathbf{k} = \begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 3 & 3 & 6 \\ 1 & -7 & -7 & 0 \\ 4 & 1 & 0 & 4 \end{pmatrix}}$$

Task 0.2 — Non-Linearity (ReLU)

After convolution, we apply a non-linear activation function. The most commonly used is **ReLU** (Rectified Linear Unit):

$$\mathrm{ReLU}(x) = \max(0,\, x)$$

This replaces every negative value with zero, keeping positives unchanged. Applied element-wise to $\mathbf{I} * \mathbf{k}$:

$$\boxed{\mathrm{ReLU}(\mathbf{I} * \mathbf{k}) = \begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 3 & 3 & 6 \\ 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 4 \end{pmatrix}}$$

(Both -7 entries at positions $(3,2)$ and $(3,3)$ become 0; all other values were already non-negative.)

Task 0.3 — Max Pooling

Max Pooling with filter (2×2) and *valid* padding divides the 4×4 feature map into non-overlapping 2×2 blocks and returns the maximum of each block, producing a 2×2 output.

$$\underbrace{\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 3 & 3 & 6 \\ 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 4 \end{pmatrix}}_{\text{ReLU output}} \longrightarrow \text{Max Pooling} \longrightarrow \begin{pmatrix} \max\{4,5,5,3\} & \max\{6,4,3,6\} \\ \max\{1,0,4,1\} & \max\{0,0,0,4\} \end{pmatrix}$$

$$\boxed{\mathbf{P} = \begin{pmatrix} 5 & 6 \\ 4 & 4 \end{pmatrix}}$$

Task 0.4 — Flattening

Flattening reshapes the matrix into a 1-D vector by reading values row by row:

$$\begin{pmatrix} 5 & 6 \\ 4 & 4 \end{pmatrix} \longrightarrow \boxed{\mathbf{v} = \begin{pmatrix} 5 & 6 & 4 & 4 \end{pmatrix}^\top}$$

Task 0.5 — Fully Connected Layer

A fully connected layer computes $\mathbf{z} = \mathbf{W}\mathbf{v}$, where $\mathbf{W}$ is a learned weight matrix. Using an example 3×4 weight matrix:

$$\mathbf{W} = \begin{pmatrix} 0.1 & 0.2 & 0.3 & 0.1 \\ 0.4 & 0.1 & 0.2 & 0.3 \\ 0.2 & 0.3 & 0.1 & 0.4 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 5 \\ 6 \\ 4 \\ 4 \end{pmatrix}$$

$$\begin{aligned} z_1 &= 0.1(5) + 0.2(6) + 0.3(4) + 0.1(4) = 0.5 + 1.2 + 1.2 + 0.4 = 3.3 \\ z_2 &= 0.4(5) + 0.1(6) + 0.2(4) + 0.3(4) = 2.0 + 0.6 + 0.8 + 1.2 = 4.6 \\ z_3 &= 0.2(5) + 0.3(6) + 0.1(4) + 0.4(4) = 1.0 + 1.8 + 0.4 + 1.6 = 4.8 \end{aligned}$$

$$\boxed{\mathbf{z} = \begin{pmatrix} 3.3 \\ 4.6 \\ 4.8 \end{pmatrix}}$$

Task 0.6 — Softmax

Softmax converts raw scores into a probability distribution:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

For $\mathbf{z} = (3.3,\ 4.6,\ 4.8)^\top$:

$$\begin{aligned} e^{3.3} \approx 27.11, \quad e^{4.6} \approx 99.48, \quad e^{4.8} \approx 121.51 \\ \text{Sum} = 27.11 + 99.48 + 121.51 = 248.10 \end{aligned}$$

$$\text{softmax}(\mathbf{z}) = \begin{pmatrix} 27.11/248.10 \\ 99.48/248.10 \\ 121.51/248.10 \end{pmatrix} \approx \begin{pmatrix} 0.109 \\ 0.401 \\ 0.490 \end{pmatrix}$$

$$\boxed{\textbf{Output class} = \text{class 3 — highest probability } \approx 0.490}$$

Task 0.7 — Loss Functions

Given:

$$\mathbf{g} = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} 0.25 & 0.6 & 0.15 \end{pmatrix}$$

The correct class is $j = 1$ (the index where $g_j = 1$).

Cross-Entropy Loss

$$H(\mathbf{y}, \mathbf{g}) = -\sum_i g_i \log(y_i) = -(0 \cdot \log 0.25 + 1 \cdot \log 0.6 + 0 \cdot \log 0.15) = -\log(0.6)$$

$$\boxed{H(\mathbf{y}, \mathbf{g}) \approx 0.511}$$

Only the term corresponding to the true class ($g_i = 1$) contributes; all others vanish. A perfect prediction of $y_1 = 1.0$ would yield $H = 0$.

Mean Squared Error Loss

$$\mathrm{MSE}(\mathbf{y}, \mathbf{g}) = \frac{1}{n}\sum_i (y_i - g_i)^2 = \frac{1}{3}\left[(0.25-0)^2 + (0.6-1)^2 + (0.15-0)^2\right]$$
$$= \frac{1}{3}[0.0625 + 0.16 + 0.0225] = \frac{0.245}{3}$$
$$\boxed{\mathrm{MSE}(\mathbf{y}, \mathbf{g}) \approx 0.082}$$

Hinge Loss (SVM Loss)

$$\mathrm{SVM}(\mathbf{y}, j) = \sum_{i \neq j} \max(0,\ y_i - y_j + 1)$$

Here $j = 1$, so $y_j = 0.6$. Summing over $i \in \{0, 2\}$:

$$i = 0: \quad \max(0,\ 0.25 - 0.6 + 1) = \max(0,\ 0.65) = 0.65$$
$$i = 2: \quad \max(0,\ 0.15 - 0.6 + 1) = \max(0,\ 0.55) = 0.55$$

$$\boxed{\mathrm{SVM}(\mathbf{y}, 1) = 0.65 + 0.55 = 1.20}$$

The hinge loss is non-zero because the correct class score does not exceed the others by a margin of 1.

Part 2: Evaluation Metrics

Task 0.1 — Theoretical Foundations

1. When is accuracy not a good idea?

Accuracy is unreliable when the dataset is *class-imbalanced*. For example, if 95% of pixels are background, a model that always predicts "background" achieves 95% accuracy while completely failing to detect any other class. Accuracy hides this failure because it treats all correct predictions equally regardless of class frequency.

2. Why is accuracy less desirable than IoU in multi-class settings?

Accuracy counts *all* correct predictions including true negatives:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

In pixel-level multi-class labeling, the number of true negatives (pixels correctly not assigned to a given class) can be enormous, artificially inflating accuracy. The Jaccard Index ignores true negatives entirely:

$$\text{IoU} = \frac{TP}{TP + FP + FN}$$

making it a stricter and more informative metric.

3. Jaccard Index vs. F1-Measure, and which is preferred for neural networks?

Both measure overlap between prediction and ground truth. The relationship is:

$$\text{IoU} = \frac{TP}{TP + FP + FN}, \qquad \text{F1} = \frac{2\,TP}{2\,TP + FP + FN}$$

F1 is always $\geq$ IoU. **F1 is generally preferred for neural networks** because it is the harmonic mean of Precision and Recall, is more sensitive to class imbalance, and is more directly connected to standard classification objectives.

Task 0.2 — Practice

Data

Position	1	2	3	4	5	6	7	8
GT	B	T	B	B	TD	TD	TD	TD
Pred	B	B	TD	BD	BC	TC	T	TD

Classes: **B** = Background, **T** = Text, **D** = Decoration, **C** = Comment. This is a multi-label scenario — a pixel may belong to more than one class simultaneously.

Definitions:

- **TP**: class present in both GT and Prediction
- **FP**: class in Prediction but not in GT
- **FN**: class in GT but not in Prediction
- **TN**: class absent from both

Class B — Background

Pos	GT has B	Pred has B	Result
1	✓	✓	TP
2	×	✓	FP
3	✓	×	FN
4	✓	✓	TP
5	×	✓	FP
6	×	×	TN
7	×	×	TN
8	×	×	TN

$TP = 2,\ FP = 2,\ FN = 1$

$$\text{Precision}_B = \frac{2}{2+2} = 0.500, \quad \text{Recall}_B = \frac{2}{2+1} = 0.667, \quad \text{F1}_B = \frac{2 \times 0.500 \times 0.667}{0.500 + 0.667} = 0.571, \quad \text{IoU}_B = \frac{2}{2+2+1} = 0$$

Class T — Text

Pos	GT has T	Pred has T	Result
1	×	×	TN
2	✓	×	FN
3	×	✓	FP
4	×	×	TN
5	✓	×	FN
6	✓	✓	TP
7	✓	✓	TP
8	✓	✓	TP

$TP = 3,\ FP = 1,\ FN = 2$

$$\text{Precision}_T = \frac{3}{4} = 0.750, \quad \text{Recall}_T = \frac{3}{5} = 0.600, \quad \text{F1}_T = \frac{2 \times 0.750 \times 0.600}{1.350} = 0.667, \quad \text{IoU}_T = \frac{3}{6} = 0.500$$

Class D — Decoration

Pos	GT has D	Pred has D	Result
1	×	×	TN
2	×	×	TN
3	×	✓	FP
4	×	✓	FP
5	✓	×	FN
6	✓	×	FN
7	✓	×	FN
8	✓	✓	TP

$TP = 1,\ FP = 2,\ FN = 3$

$$\text{Precision}_D = \frac{1}{3} = 0.333, \quad \text{Recall}_D = \frac{1}{4} = 0.250, \quad \text{F1}_D = \frac{2 \times 0.333 \times 0.250}{0.583} = 0.286, \quad \text{IoU}_D = \frac{1}{6} = 0.167$$

Class C — Comment

Pos	GT has C	Pred has C	Result
1–4	×	×	TN
5	×	✓	FP
6	×	✓	FP
7–8	×	×	TN

$TP = 0,\ FP = 2,\ FN = 0.$ Class C never appears in the ground truth.

$$\text{Precision}_C = 0.000, \quad \text{Recall}_C = \text{undefined} \to 0.000, \quad \text{F1}_C = 0.000, \quad \text{IoU}_C = 0.000$$

Summary Table

Class	**GT Freq.**	**Precision**	**Recall**	**F1**	**IoU**
B	3	0.500	0.667	0.571	0.400
T	5	0.750	0.600	0.667	0.500
D	4	0.333	0.250	0.286	0.167
C	0	0.000	0.000	0.000	0.000

GT Frequency = number of pixels in which the class appears in the ground truth.

Mean — Macro (unweighted, divided by number of classes $= 4$**)**

$$\text{Macro Precision} = \frac{0.500 + 0.750 + 0.333 + 0.000}{4} = \mathbf{0.396}$$
$$\text{Macro Recall} = \frac{0.667 + 0.600 + 0.250 + 0.000}{4} = \mathbf{0.379}$$
$$\text{Macro F1} = \frac{0.571 + 0.667 + 0.286 + 0.000}{4} = \mathbf{0.381}$$
$$\text{Macro IoU} = \frac{0.400 + 0.500 + 0.167 + 0.000}{4} = \mathbf{0.267}$$

Mean — Weighted by Class Frequency (total GT labels $= 3 + 5 + 4 + 0 = 12$**)**

$$\text{Weighted Precision} = \frac{3(0.500) + 5(0.750) + 4(0.333) + 0(0.000)}{12} = \frac{1.500 + 3.750 + 1.333}{12} = \frac{6.583}{12} = \mathbf{0.549}$$
$$\text{Weighted Recall} = \frac{3(0.667) + 5(0.600) + 4(0.250) + 0}{12} = \frac{2.000 + 3.000 + 1.000}{12} = \frac{6.000}{12} = \mathbf{0.500}$$
$$\text{Weighted F1} = \frac{3(0.571) + 5(0.667) + 4(0.286) + 0}{12} = \frac{1.714 + 3.333 + 1.143}{12} = \frac{6.190}{12} = \mathbf{0.516}$$
$$\text{Weighted IoU} = \frac{3(0.400) + 5(0.500) + 4(0.167) + 0}{12} = \frac{1.200 + 2.500 + 0.667}{12} = \frac{4.367}{12} = \mathbf{0.364}$$